# ITA0506-COMPUTER VISION   LAB EXPERIMENT

## DATE :09-07-2026

## EXPNO:1

## AIM: Perform basic Image Handling and processing operations on the image.

## • Read an image in python and Convert an Image to Grayscale

## Algorithm:

**Steps: Convert Image to Grayscale**

1. **Import the OpenCV library**
   Load cv2 to perform image processing.
2. **Read the image**
   Use cv2.imread() to load the image from the file path.
3. **Check the image**
   Ensure the image is loaded properly (img != None).
4. **Convert to grayscale**
   Use cv2.cvtColor(img, cv2.COLOR_BGR2GRAY).
5. **Display or save the output**
   Show using cv2.imshow() or save using cv2.imwrite().

## Code:

EXP NO 1COMPUTER VISON.py - C:\Users\surya\Downloads\EXP NO 1COMPUTER VISON.py (3.13.5)

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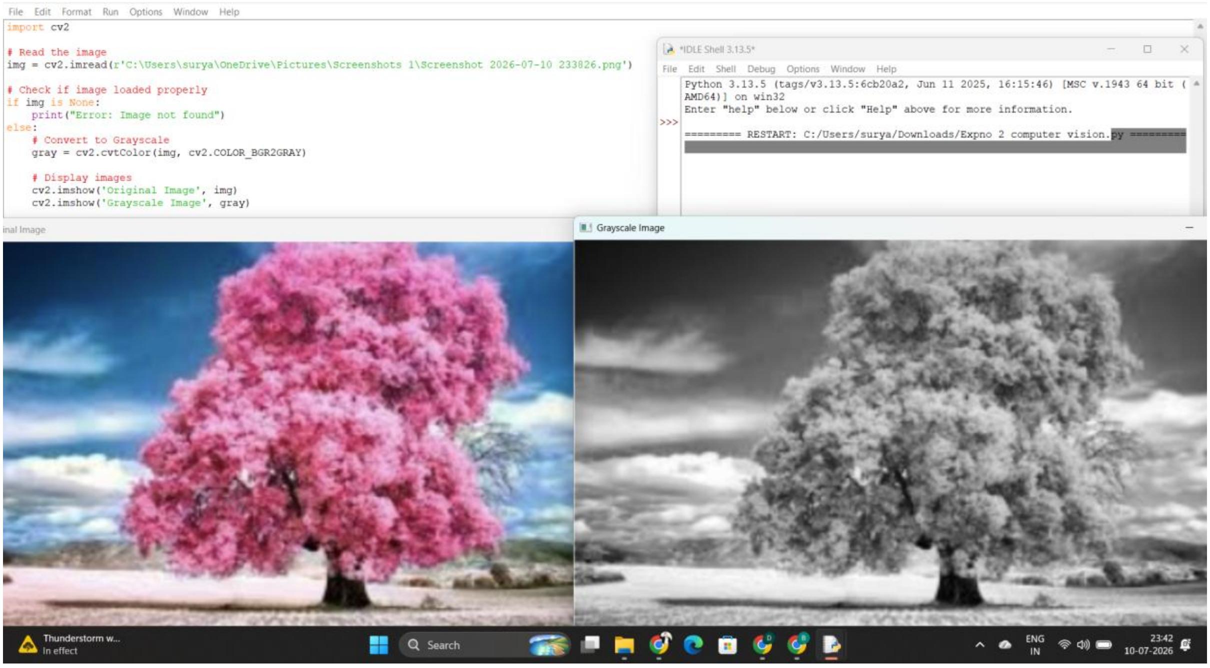
```
import cv2

# Read image (fixed path)
img = cv2.imread(r'C:\Users\surya\OneDrive\Pictures\Screenshots 1\Screenshot 2026-07-10 232758.png')

# Check if image loaded
if img is None:
    print("Error: Image not found")
else:
    # Apply Gaussian Blur
    blur = cv2.GaussianBlur(img, (15, 15), 0)

    # Show images
    cv2.imshow('Original Image', img)
    cv2.imshow('Blurred Image', blur)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

# OUTPUT :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import cv2
# Read the image
img = cv2.imread(r'C:\Users\surya\OneDrive\Pictures\Screenshots 1\Screenshot 2026-07-10 233826.png')
# Check if image loaded properly
if img is None:
print("Error: Image not found")
else:
# Convert to Grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
# Display images
cv2.imshow('Original Image', img)
cv2.imshow('Grayscale Image', gray)
*IDLE Shell 3.13.5*
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========= RESTART: C:/Users/surya/Downloads/Expno 2 computer vision.py =========
Grayscale Image
Thunderstorm w...
In effect
Search
ENG
IN
23:42
10-07-2026